ETL Transformation Example: Sorting and Finalizing the Data

Explore how to execute the transformation step within an ETL pipeline by sorting raw data by date in descending order. Learn to implement a shell script that cleans and formats the data for subsequent loading, ensuring the dataset meets analytic needs before moving forward.

Task 5: Sort the data

Finally, the last task is to sort the data by date in descending order.

transform_data_5... Shell

```shell
#!/bin/bash

echo -e "\nTask #5 - Sorting the Data"
sort_data(){

    # Sort the data by date
    cat raw_data.csv | sort -t ","  -k1.7,1.10 -k1.1,1.2 -k1.4,1.5 -r >>

    echo "Done"
}
sort_data
```

Ask

Sorting the data by date: transform_data_5_sort.sh

We've created a file called `transform_data_5_sort.sh` to perform these operations. Let's walk through the code.

- **Line 7**:
  - We print the contents of `raw_data.csv`, sort the contents of `raw_data.csv`, and specify that commas separate the file.
  - We sort by three values:
    - Year(`-k1.7,1.10`): First column and the characters between locations 7 to 10
    - Month(`-k1.1,1.2`): First column and the characters between locations 1 to 2
    - Day(`-k1.4,1.5`): First column and the characters between locations 4 to 5
  - Then, we append the results to a new file called `clean_data.csv`.

Next, we need to add `transform_data_5_sort.sh` to the pipeline, and we are done. Run the entire Extract and Transform steps from start to finish by clicking on the code widget below.

Files

main.sh
clean_d...
transfor...
transfor...
transfor...
raw_dat...
transfor...
extract_...
transfor...

main.sh Shell

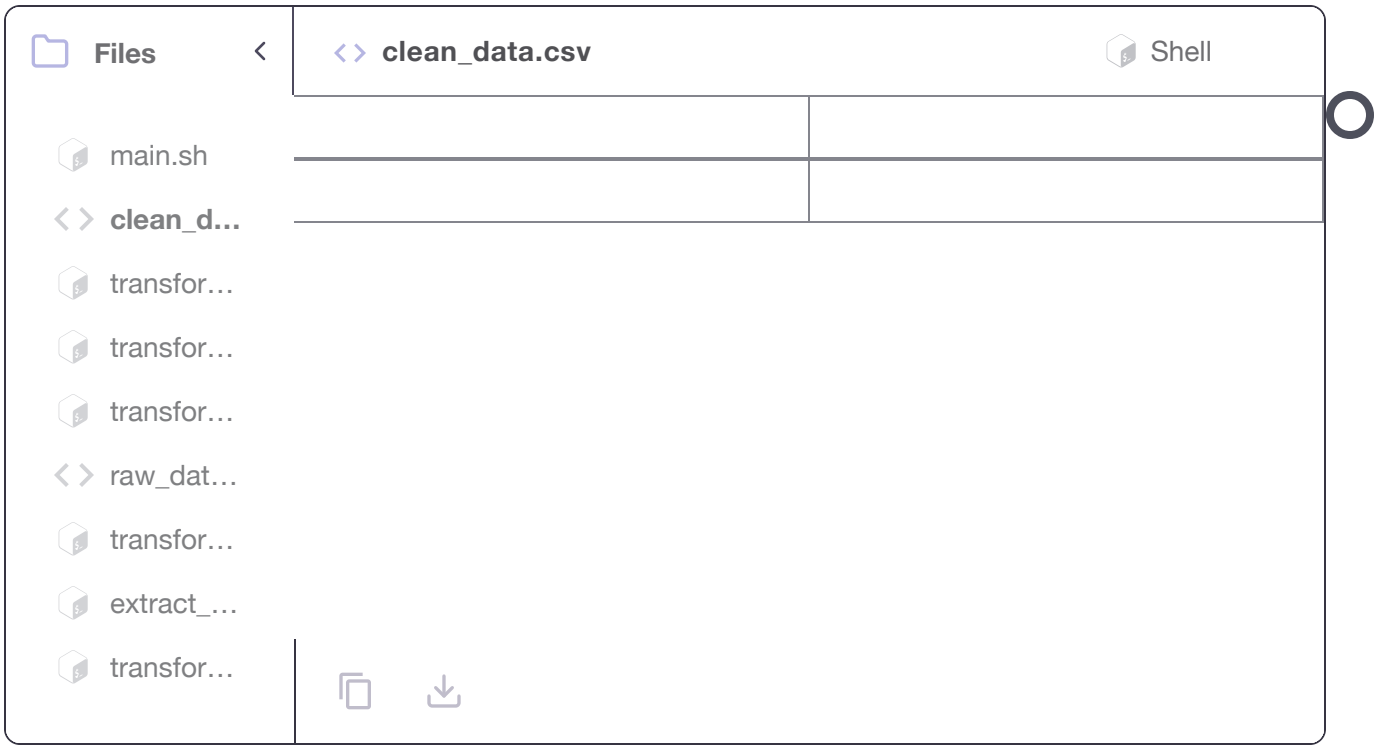
```
#!/bin/bash

# Setting the project location as an environment variabl
export WORKDIR='/usercode/ETL_Lottery'

# Changing directories to the project folder
cd $WORKDIR

# Extract Raw Data
bash extract_data.sh

# Issue #1 - Make the Data Tabluar
bash transform_data_1_tabulate.sh
```

Ask

Run

ETL pipeline: main.sh

Before moving on to the load step in the pipeline, let's take a final look at the new and clean file called `clean_data.csv`.



clean_data.csv

The transform step in the ETL pipeline is done. The file has been cleaned, sorted, and transformed according to the requested format and needs of the data scientist.

Mark As Completed

Ask